



CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

9702/22

October/November 2021

1 hour 15 minutes

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
	$(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas

$$W = p\Delta V$$

gravitational potential

$$\phi = -\frac{Gm}{r}$$

hydrostatic pressure

$$p = \rho gh$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

Doppler effect

$$f_o = \frac{f_s v}{v \pm v_s}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor

$$W = \frac{1}{2} QV$$

electric current

$$I = Anvq$$

resistors in series

$$R = R_1 + R_2 + \dots$$

resistors in parallel

$$1/R = 1/R_1 + 1/R_2 + \dots$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 \exp(-\lambda t)$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

Answer **all** the questions in the spaces provided.

- 1 (a) A unit may be stated with a prefix that represents a power-of-ten multiple or submultiple.

Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power-of-ten multiple or submultiple.

Table 1.1

prefix	power-of-ten multiple or submultiple
kilo (k)	10^3
tera (T)	
()	10^{-12}

[2]

- (b) In the following list, underline all the units that are SI base units.

ampere

coulomb

metre

newton

[1]

- (c) The potential difference V between the two ends of a uniform metal wire is given by

$$V = \frac{4\rho LI}{\pi d^2}$$

where d is the diameter of the wire,
 I is the current in the wire,
 L is the length of the wire,
 and ρ is the resistivity of the metal.

For a particular wire, the percentage uncertainties in the values of some of the above quantities are listed in Table 1.2.

Table 1.2

quantity	percentage uncertainty
d	$\pm 3.0\%$
I	$\pm 2.0\%$
L	$\pm 2.5\%$
V	$\pm 3.5\%$

The quantities listed in Table 1.2 have values that are used to calculate ρ as $4.1 \times 10^{-7} \Omega \text{ m}$.

For this value of ρ , calculate:

- (i) the percentage uncertainty

percentage uncertainty =% [2]

- (ii) the absolute uncertainty.

absolute uncertainty = $\Omega \text{ m}$ [1]

[Total: 6]

- 2 A charged oil drop is in a vacuum between two horizontal metal plates. A uniform electric field is produced between the plates by applying a potential difference of 1340V across them, as shown in Fig. 2.1.

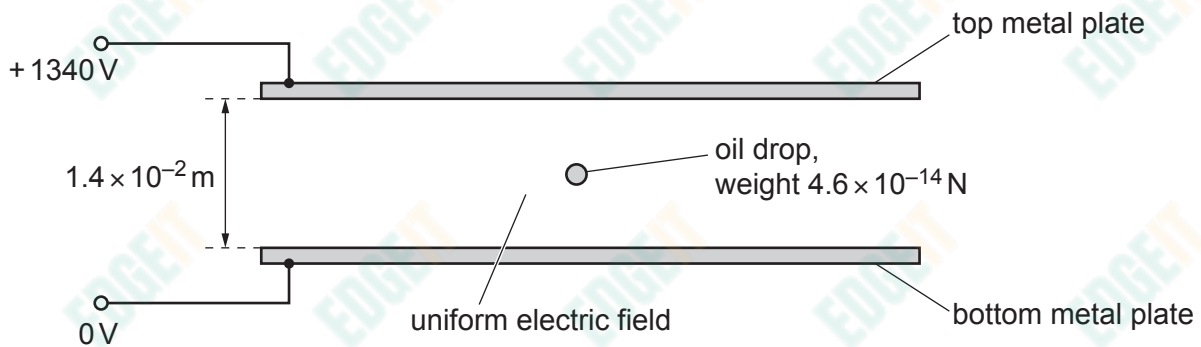


Fig. 2.1

The separation of the plates is $1.4 \times 10^{-2} \text{ m}$.

The oil drop of weight $4.6 \times 10^{-14} \text{ N}$ remains stationary at a point mid-way between the plates.

- (a) (i) Calculate the magnitude of the electric field strength.

electric field strength = NC^{-1} [2]

- (ii) Determine the magnitude and the sign of the charge on the oil drop.

magnitude of charge = C

sign of charge [3]

- (b) The electric potentials of the plates are instantaneously reversed so that the top plate is at a potential of 0V and the bottom plate is at a potential of +1340V. This change causes the oil drop to start moving downwards.

- (i) Compare the new pattern of the electric field lines between the plates with the original pattern.

.....
 [2]

- (ii) Determine the magnitude of the resultant force acting on the oil drop.

resultant force = N [1]

- (iii) Show that the magnitude of the acceleration of the oil drop is 20 ms^{-2} .

[2]

- (iv) Assume that the radius of the oil drop is negligible.

Use the information in **(b)(iii)** to calculate the time taken for the oil drop to move to the bottom metal plate from its initial position mid-way between the plates.

time = s [2]

- (c) The oil drop in **(b)** starts to move at time $t = 0$. The distance of the oil drop from the bottom plate is x .

On Fig. 2.2, sketch the variation with time t of distance x for the movement of the drop from its initial position until it hits the surface of the bottom plate. Numerical values of t are not required.

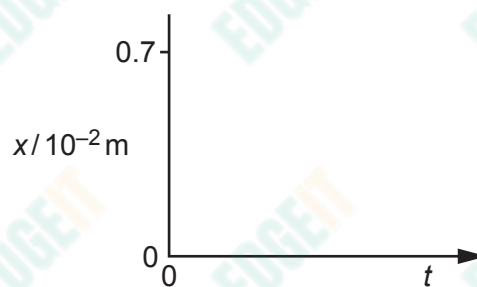


Fig. 2.2

[2]

[Total: 14]

- 3 (a) Define *power*.

.....
 [1]

- (b) A car of mass 1700 kg moves in a straight line along a slope that is at an angle θ to the horizontal, as shown in Fig. 3.1.

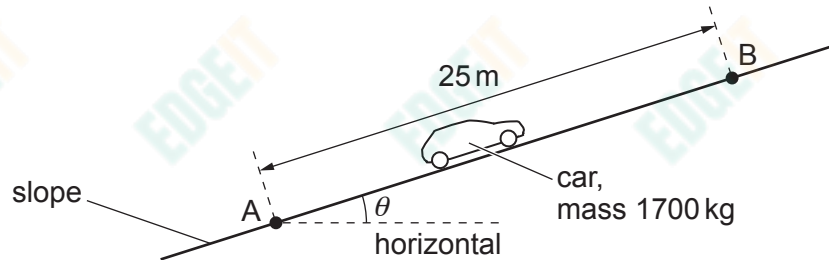


Fig. 3.1 (not to scale)

The car moves at constant velocity for a distance of 25 m from point A to point B. Air resistance and friction provide a total resistive force of 440 N that opposes the motion of the car.

For the movement of the car from A to B:

- (i) state the change in the kinetic energy

change in kinetic energy = J [1]

- (ii) calculate the work done against the total resistive force.

work done = J [1]

- (c) The movement of the car in (b) from A to B causes its gravitational potential energy to increase by $4.8 \times 10^4 \text{ J}$.

Calculate:

- (i) the increase in vertical height h of the car for its movement from A to B

$$h = \dots\dots\dots \text{ m [2]}$$

- (ii) angle θ .

$$\theta = \dots\dots\dots^\circ \text{ [1]}$$

- (d) The engine of the car in (b) produces an output power of $1.7 \times 10^4 \text{ W}$ to move the car along the slope.

Calculate the time taken for the car to move from A to B.

$$\text{time} = \dots\dots\dots \text{ s [2]}$$

[Total: 8]

- 4 A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

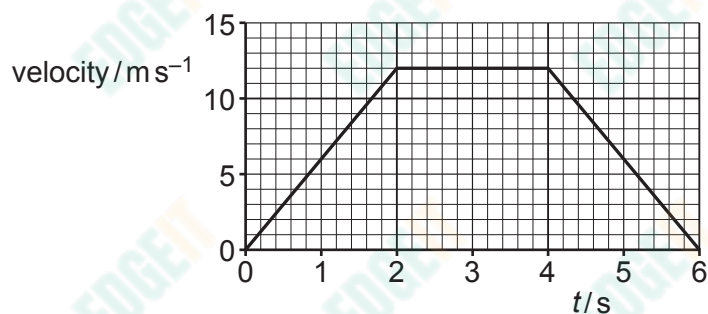


Fig. 4.1

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

- (i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

- (ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

- (c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

[Total: 6]

- 5 A tube is initially fully submerged in water. The axis of the tube is kept vertical as the tube is slowly raised out of the water, as shown in Fig. 5.1.

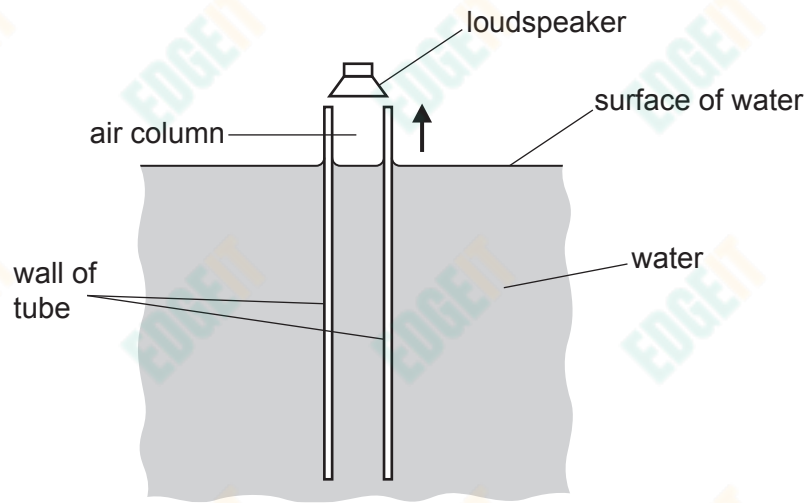


Fig. 5.1

A loudspeaker producing sound of frequency 530 Hz is positioned at the open top end of the tube as it is raised. The water surface inside the tube is always level with the water surface outside the tube. The speed of the sound in the air column in the tube is 340 m s^{-1} .

- (a) Describe a simple way that a student, without requiring any additional equipment, can detect when a stationary wave is formed in the air column as the tube is being raised.

.....
 [1]

- (b) Determine the height of the top end of the tube above the surface of the water when a stationary wave is first produced in the tube. Assume that an antinode is formed level with the top of the tube.

height = m [3]

- (c) Determine the distance moved by the tube between the positions at which the first and second stationary waves are formed.

distance = m [1]

[Total: 5]

- 6 A cell of electromotive force (e.m.f.) 0.48 V is connected to a metal wire X, as shown in Fig. 6.1.

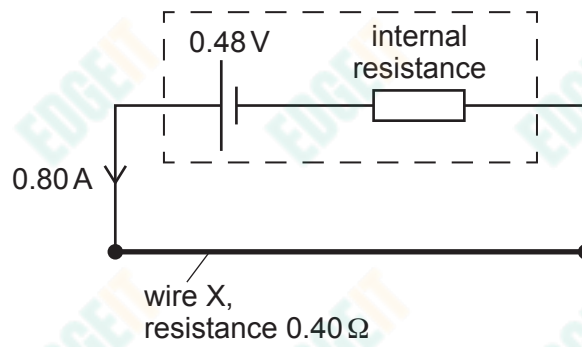


Fig. 6.1

The cell has internal resistance. The current in the cell is 0.80 A .

Wire X has length 3.0 m , cross-sectional area $1.3 \times 10^{-7}\text{ m}^2$ and resistance $0.40\ \Omega$.

- (a) Calculate the charge passing through the cell in a time of 7.5 minutes.

charge = C [2]

- (b) Calculate the percentage efficiency with which the cell supplies power to wire X.

efficiency = % [3]

- (c) There are 3.2×10^{22} free (conduction) electrons contained in the volume of wire X.

For wire X, calculate:

- (i) the number density n of the free electrons

$$n = \dots\dots\dots \text{m}^{-3} \quad [1]$$

- (ii) the average drift speed of the free electrons.

$$\text{average drift speed} = \dots\dots\dots \text{ms}^{-1} \quad [2]$$

- (d) A wire Y has the same cross-sectional area as wire X and is made of the same metal. Wire Y is longer than wire X.

Wire X in the circuit is now replaced by wire Y. Assume that wire Y has the same temperature as wire X.

State and explain whether the average drift speed of the free electrons in wire Y is greater than, the same as, or less than that in wire X.

.....

.....

.....

.....

.....

..... [3]

[Total: 11]

- 7 A stationary nucleus P of mass 243u decays by emitting an α -particle of mass 4u to form a different nucleus Q, as illustrated in Fig. 7.1.

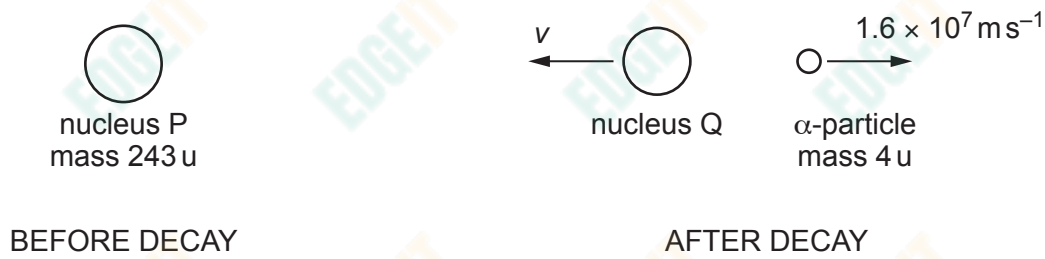


Fig. 7.1

The initial speed of the α -particle is $1.6 \times 10^7 \text{ms}^{-1}$.

- (a) Use the principle of conservation of momentum to explain why the initial velocities of nucleus Q and the α -particle must be in opposite directions.

.....

.....

.....

..... [2]

- (b) Determine the initial speed v of nucleus Q.

$v = \dots\dots\dots \text{ms}^{-1}$ [2]

- (c) Calculate the initial kinetic energy, in MeV, of the α -particle.

kinetic energy = $\dots\dots\dots \text{MeV}$ [3]

(d) A graph of number of neutrons N against proton number Z is shown in Fig. 7.2.

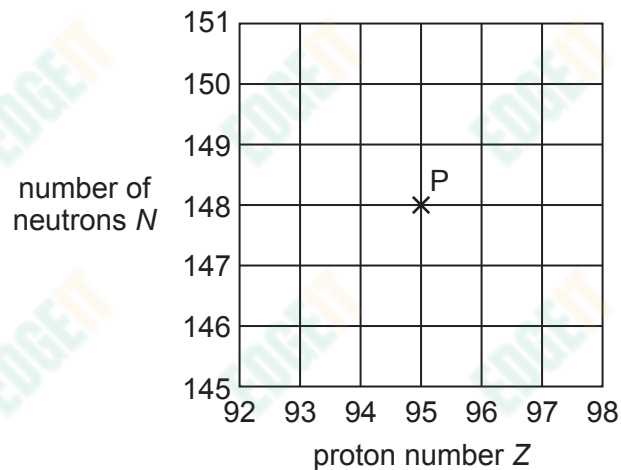


Fig. 7.2

The graph shows a cross that represents nucleus P.

A nucleus R has a nucleon number of 242 and is an isotope of nucleus P.

Nucleus R decays by emitting a β^- particle to form a different nucleus S.

(i) On Fig. 7.2, draw a cross to represent:

1. nucleus R (label this cross R)
2. nucleus S (label this cross S).

[2]

(ii) State the name of the other lepton, in addition to the β^- particle, that is emitted during the decay of nucleus R.

..... [1]

[Total: 10]

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.